Higher A · Cell Biology

40 marks • About 45 minutes • Scientific calculator permitted.

Original Study Hub review paper for AQA Trilogy (8464), not an official AQA paper. Answer every part; show working. For “Choose one”, circle one letter. Answer spaces are indicative. Marking guides are separate.

1. Cells and their structures · 5 marks

1a [1 mark]

Which structure-function pair is correct? Choose one.

A. Mitochondria - protein synthesis

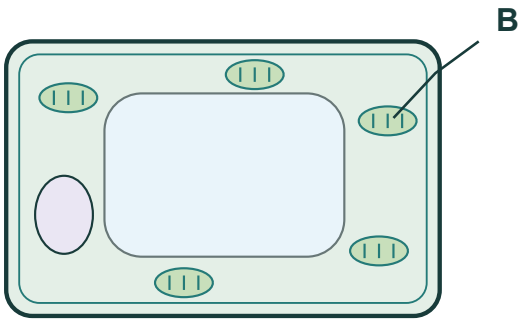
B. Nucleus - aerobic respiration

C. Ribosomes - protein synthesis

1b [2 marks]

Figure H-CELL shows a leaf cell. Name structure B. State its function.

H-CELL · Leaf cell



Leaf cell • schematic, not to scale

1c [1 mark]

Which description fits the main genetic material of a typical bacterial cell? Choose one.

A. A DNA loop not enclosed by a nucleus

B. Chromosome pairs enclosed by a nucleus

C. DNA found only inside mitochondria

1d [1 mark]

A bacterial cell is 3 micrometres long and an animal cell is 30 micrometres long. How many orders of magnitude separate these lengths? Choose one.

A. Two

B. Ten

C. One

2. Microscopy and RP01 · 6 marks

2a [1 mark]

Which procedure is suitable when changing from low to high power after locating a specimen? Choose one.

A. Use coarse focusing rather than fine focusing after changing to high power

B. Centre first, then use fine focus

C. Move the specimen out of view before changing power

2b [1 mark]

Explain what higher resolving power allows a microscope to do.

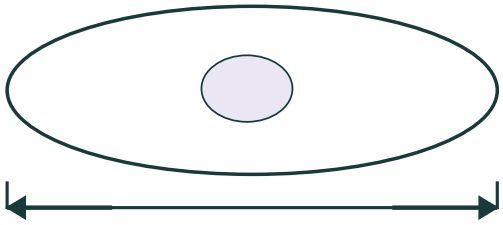
2c [3 marks]

The recorded image length is 72 mm and the magnification is ×24 000. Calculate the real cell length in metres. Give the answer in standard form and show working.

2d [1 mark]

A student writes “actual cell length = 4 cm” by measuring a magnified drawing on a screen. Explain one change needed for a valid size record.

H-DRAWING · Student record



Measured on screen: 4 cm

Student note: “Actual cell length = 4 cm”

3. Cell division and specialisation · 6 marks

3a [1 mark]

Which statement about body-cell genetic material is correct? Choose one.

A. Chromosomes normally occur in pairs and each carries many genes

B. All genes occur on a single chromosome in every body cell

C. Chromosomes are produced only after the cytoplasm divides

3b [2 marks]

Explain why the two new nuclei formed by mitosis normally contain the same genes as the original nucleus.

3c [1 mark]

Which comparison of plant and animal differentiation is correct? Choose one.

A. Differentiation and mitosis always mean the same process

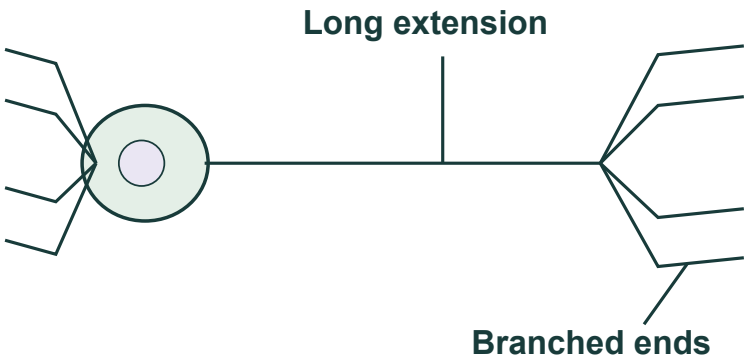
B. Many plant cells retain differentiation capacity throughout life; most animal types differentiate early

C. Most animal cells retain unlimited differentiation capacity, but plant cells never differentiate

3d [2 marks]

Figure H-NERVE shows a cell transmitting signals from the spinal cord to a foot. It has a long extension and branched ends. Explain how both features help its function.

H-NERVE · Nerve cell



Schematic, not to scale

4. Stem cells and evidence · 5 marks

4a [2 marks]

Compare the range of cells that embryonic stem cells and plant meristem cells can produce.

4b [1 mark]

Why are stem cells from an embryo produced by therapeutic cloning less likely to be rejected by the patient? Choose one.

A. Differentiation removes all genetic differences between donors

B. The embryo has the same genes as the patient

C. All stem cells from every donor already have identical genes

4c [2 marks]

Use H-STEM-TABLE. A nursery claims that meristem cloning guarantees the long-term survival of this rare plant. Evaluate the claim using one evidence-based benefit and one limitation.

H-STEM-TABLE · Rare-plant cloning

H-STEM-TABLE · Rare-plant cloning

Feature	Information
Production	Many identical plants produced quickly
Existing trait	Current disease resistance is preserved
Concern	A new disease may affect many genetically uniform clones

5. Diffusion and exchange · 6 marks

5a [1 mark]

Which statement describes net diffusion at a concentration gradient? Choose one.

A. No particles can move against the gradient individually

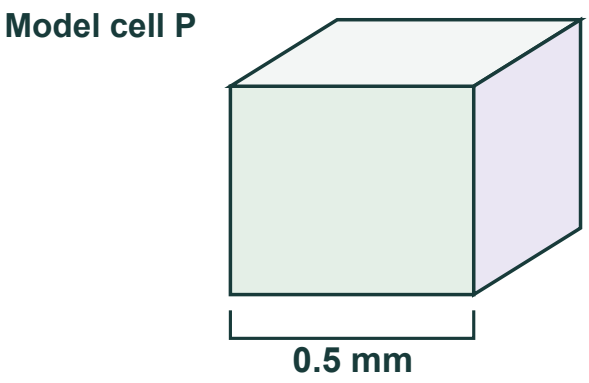
B. Respiration drives every particle crossing a cell membrane

C. More particles move down the gradient than up it overall

5b [2 marks]

Model cell P is a cube with side length 0.5 mm. Model cell Q has surface-area-to-volume ratio 3:1, measured using mm. Calculate P’s ratio and state how many times as large as Q’s it is.

H-CUBE · Cube model



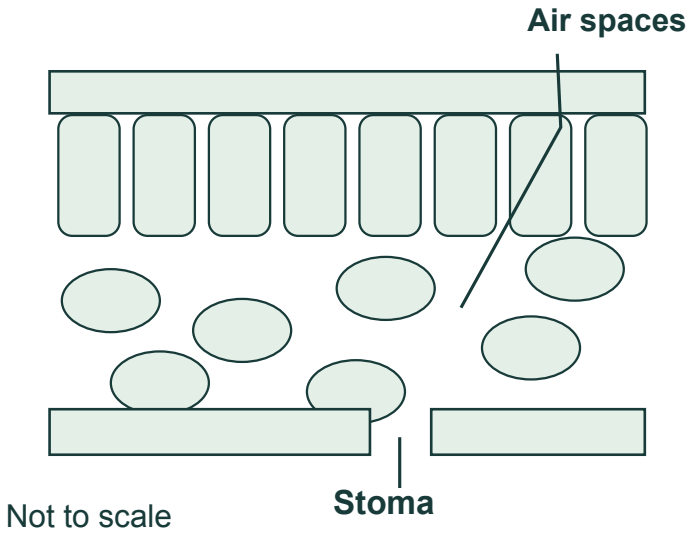
5. Diffusion and exchange · continued

5c [2 marks]

Figure H-LEAF shows a thin leaf with many air spaces and stomata. Explain how two features make diffusion of carbon dioxide to photosynthesising cells effective.

H-LEAF · Leaf exchange surface

Leaf section



5d [1 mark]

The table gives diffusion rates at 15 °C: 0.24, 0.26, 0.25 units/min; at 25 °C: 0.44, 0.46, 0.45 units/min, with other conditions controlled. Which conclusion is supported? Choose one.

H-DIFF-TABLE · Rates / units/min

H-DIFF-TABLE · Rates / units/min

Temp / °C	Trial 1	Trial 2	Trial 3
15	0.24	0.26	0.25
25	0.44	0.46	0.45

A. The measurements prove the rate doubles for every future temperature rise

B. Diffusion stops at the lower temperature

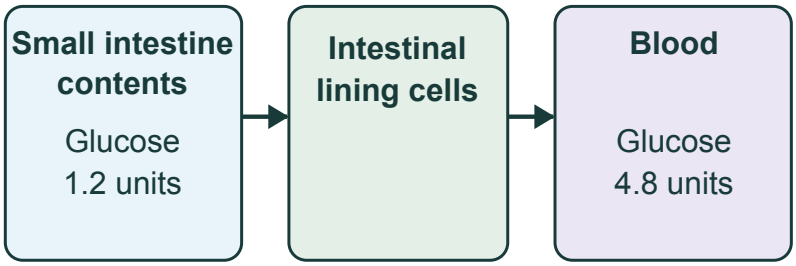
C. The higher-temperature measurements consistently show a faster rate under these conditions

6. Transport processes · 4 marks

6a [2 marks]

H-GLUCOSE shows glucose uptake from the small intestine into blood. A treatment reduces respiration in the intestinal lining cells; the concentration difference initially stays unchanged. Predict the effect on continued uptake and explain why.

H-GLUCOSE · Glucose uptake route



Same arbitrary concentration units

Overall route through cells; not one direct pump

6b [1 mark]

Which statement defines osmosis completely? Choose one.

A. Net movement of water from a concentrated to a dilute solution through a partially permeable membrane

B. Net movement of dissolved solute from a dilute to a concentrated solution through a partially permeable membrane

C. Net movement of water from a dilute to a concentrated solution through a partially permeable membrane

6c [1 mark]

Which combination of changes increases diffusion rate when other conditions remain unchanged? Choose one.

A. Smaller exchange area and lower temperature

B. Shallower concentration gradient and lower temperature

C. Larger exchange area and steeper concentration gradient

7. RP02: mass, graphs and evaluation · 8 marks

7a [2 marks]

This is a separate pilot batch, not the tissue used for 7b. Use H-MASS-TABLE to calculate the percentage change in mass. Give your answer to one decimal place, including its sign, and show working.

H-MASS-TABLE · Pilot batch for 7a

H-MASS-TABLE · Pilot batch for 7a

Measurement	Recorded value
Sucrose (mol/dm³)	0.50
Initial mass / g	3.20
Final mass / g	2.92

7. RP02: mass, graphs and evaluation · continued

7b [6 marks]

Use H-OSM-TABLE and H-OSM-GRAPH. The investigator used one cylinder per concentration, did not blot before final weighing, and varied soaking time. They claim that every type of plant tissue has no net mass change at exactly the same sucrose concentration. Evaluate the claim. Use an estimate from the graph and osmosis, and propose an improved investigation.

H-OSM-TABLE · Investigation for 7b

H-OSM-TABLE · Investigation for 7b

Sucrose (mol/dm³)	Mass change / %
0.00	+11
0.15	+5
0.30	+1
0.45	−7.5
0.60	−12

One cylinder at each concentration

H-OSM-GRAPH · Investigation for 7b

